Character Chess Engine (CCE 1)

Skill-based and personality-driven LLM chess move selection with elo from 400 to 2000

Problem

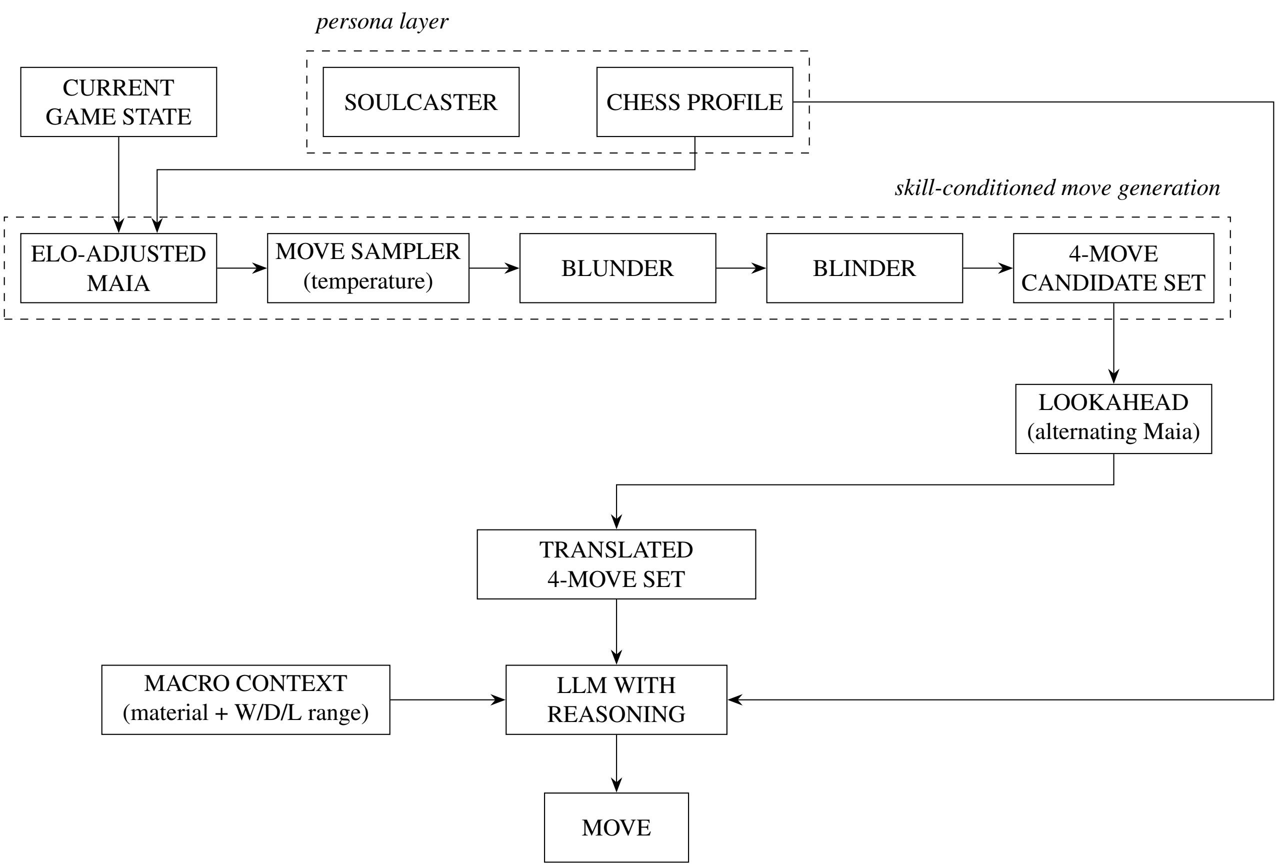
AI game companions must play chess as a believable character and as a believable player at a target rating. Standard engine weakening returns one skill-adjusted move. Our system has a second decision layer. An LLM chooses the move that fits its personality. It therefore needs a set of candidate moves, not one move. Every candidate in that set must already reflect the persona's rating. A 400-rated character and an 1800-rated character should consider different moves, imagine different continuations, and hold different beliefs about who is winning. If any layer leaks engine strength, the character breaks.

Existing Solutions

Stockfish weakens play through Skill Level and UCI_Elo. Both run a full-strength search and perturb only the final move selection. The MultiPV list stays engine-optimal at every setting, so it cannot serve as a weak player's consideration set. UCI_Elo also bottoms out near 1320. The mechanism is structurally unable to hang a piece, because the search always sees the threat.

Maia is a neural network trained on millions of human games. It predicts the probability of each legal move for a player of a given rating. Maia-2 unifies this into one model conditioned on an Elo input (1100 to 1900). It is the most accurate predictor of human moves, but it has two known gaps. Playing its argmax move yields play far above the label, since averaging many players removes their individual blunders. Its tactical sharpness also barely decreases at lower ratings. We build on Maia and correct both gaps with the sampling layers below.

Architecture



Components

- **SOULCASTER.** Generates a chess profile from a natural-language persona description. It maps character traits to numeric parameters.
- **CHESS PROFILE.** The persona config: Elo, temperature, style weights, blunder rate, blinder rate, lookahead depth, and evaluation range width. All constants are tuned empirically. Each persona plays 100 games against Stockfish UCI_Elo anchors, and its measured performance rating becomes its public label.
- **ELO-ADJUSTED MAIA.** Maia-2 inference on the current position, conditioned on persona Elo. Sub-1100 personas clamp to the 1100 floor. We set elo_oppo equal to elo_self. Output: a probability distribution $p(m)$ over all legal moves.
- **MOVE SAMPLER / TEMPERATURE.** Samples $k = 4$ moves without replacement from the tempered distribution $q(m) \propto p(m)^{1/T}$ via Gumbel-top-k: score each legal move by $s(m) = \log p(m)/T + g_m$ with $g_m \sim \mathrm{Gumbel}(0,1)$ and keep the top 4. Temperature is 1 inside Maia's calibrated range, where distribution entropy already encodes picker skill. Below the floor it extends the range down to 400: $T = 1 + \max\left(0, \frac{1100-\mathrm{Elo}}{700}\right)$, so a 400 persona gets $T = 2$. Before tempering, drop moves below 1 percent raw probability so flattening spreads mass over bad-but-human moves rather than the full legal-move tail.
- **BLUNDER.** With probability $P_{\mathrm{blunder}} = b_{\max} e^{-(\mathrm{Elo}-400)/\lambda}$, swaps one candidate for a move from the bottom tail of the distribution. Models impulsive lapses. Near certainty at 400, near zero at 2000.
- **BLINDER.** When Stockfish detects a short forced mate or tactic, removes it from the candidate set with probability $P_{\mathrm{blind}} = 1 - \sigma\left(\frac{\mathrm{Elo}-\mu}{s}\right)$. Models not seeing what is on the board, and counters Maia's tactical-sharpness quirk.
- **CURRENT GAME STATE.** The board position, side to move, and move history, fed to Maia at every turn.
- **LOOKAHEAD.** For each candidate, one rollout of alternating Maia predictions, all plies conditioned on persona Elo and sampled at persona temperature. The imagined line is rating-appropriate on both sides. One rollout per candidate, never averaged. Depth is $n(\mathrm{Elo}) = \mathrm{clip}\left(\lfloor \mathrm{Elo}/250 \rfloor, 1, 8\right)$ plies: a 400 sees only its own move, a 2000 sees four moves per side.
- **TRANSLATED 4-MOVE SET.** A deterministic python-chess tagger converts each surviving candidate and its rollout into plain sentences: captures, checks, threats, material deltas, king exposure. No raw notation reaches the LLM.
- **MACRO CONTEXT.** Material count plus a win/draw/loss assessment shown as a probability range, not a confidence score. The shown band is $[\hat{w} - r,\ \hat{w} + r]$ with $r = r_{\max}\left(1 - \frac{\mathrm{Elo}-400}{1600}\right)$, narrowing with Elo: a 400 sees 0 to 100 percent, a 2000 sees a 5-point band. At low Elo the band center also shifts toward a naive material-only evaluation, since weak players are confidently wrong, not just uncertain.
- **LLM WITH REASONING.** Receives the persona text, macro context, and the 4 translated candidates. It picks purely by character. Strength is fully determined upstream, so the LLM can only express style.

Limitations

The in-character half of the system is still weak. Translation is currently literal, move-by-move notation to sentences. It does not surface the strategic texture of a line, such as whether it is aggressive, simplifying, cramping, or risky. Without those tags the LLM cannot reliably select by playstyle, so personality expression is shallower than strength expression. The fix is a richer feature layer over each rollout before the LLM. Other known limits: everything below Elo 1100 is extrapolation, calibrated only by measured performance rather than by training data. BLINDER is a heuristic patch over Maia's tactics quirk, not a principled model of oversight. Latency grows with 4 candidates times n rollout plies of Maia inference per move. Finally, elo_oppo is fixed to the persona's own rating, so the engine does not model its actual opponent.